Time-Step-Free Physics-based Modeling of Retention Loss in Cryogenic 3-D NAND Flash

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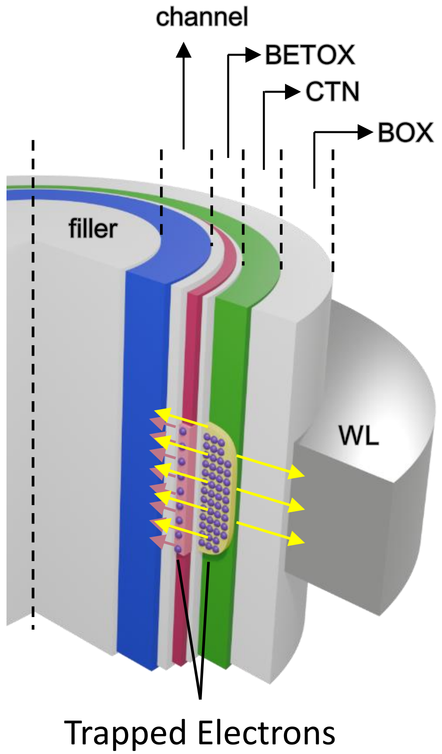
Introduction

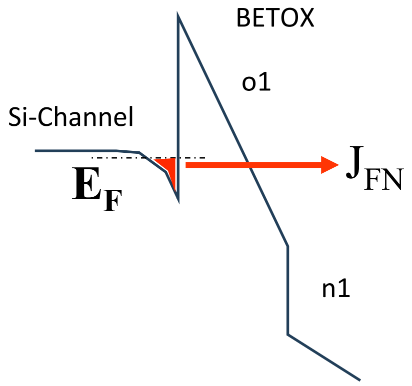


- 3-D NAND Flash memory는 charge-trap stack 내부에 저장된 trapped electron을 통해 정보를 저장함
- Retention 시간이 증가하면 trapped electron의 detrapping으로 인해 V_{th} 감소 발생
- Cylindrical 구조와 CTN 내부 arbitrary trapped electron distribution을 고려하기 위한 physics-based modeling 필요

Jin, TED 2025

BETOX Trapped Electron Detrapping Modeling

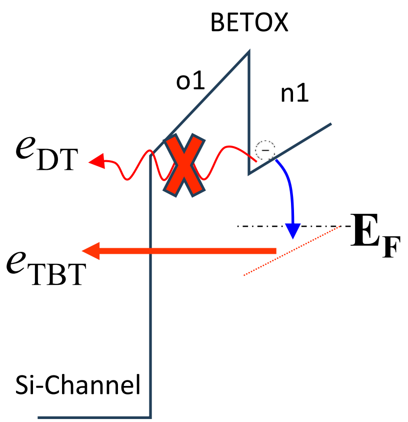
Program ($V_G = V_{pgm}$)



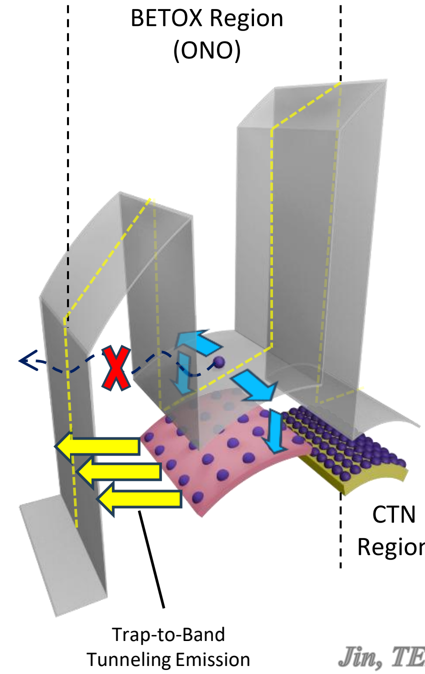
Program 동작에서는 강한 전기장에 의해 FN tunneling이 발생함

하지만, Retention 상태에서는 $V_G = 0V$ 이므로 충분한 전자가 conduction band 에 축적되지 않아 direct tunneling의 영향이 감소함

Retention ($V_G = 0$ V)



BETOX 내부에 trapped된 electron은 channel conduction band 으로 trap-to-band tunneling emission 될 수 있음



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BETOX Trapped Electron Detrapping Modeling

BETOX의 O/N/O/N/O 5-layer 구조 Poisson solution

$$V_{O1}(r) = \frac{K}{\varepsilon_{ox}} \ln \frac{r}{r_0}$$
$$V_{N1}(r) = \frac{K}{\varepsilon_{ox}} \ln \frac{r_{O1}}{r_0} + \frac{K_1}{\varepsilon_n} \ln \frac{r}{r_{O1}} + \frac{q n_{t\_tn1}}{4\varepsilon_n} (r^2 - r_{O1}^2)$$
$$V_{O2}(r) = V_{N1}(r_{n1}) + \frac{K_2}{\varepsilon_{ox}} \ln \frac{r}{r_{n1}}$$
$$V_{CTN}(r) = V_{O2}(r_{tox}) + \frac{K_3}{\varepsilon_n} \ln \frac{r}{r_{tox}} + \frac{q n_{t\_ctn}}{4\varepsilon_n} (r^2 - r_{tox}^2)$$
$$V_{box}(r) = V_{CTN}(r_{CTN}) + \frac{K_4}{\varepsilon_{ox}} \ln \frac{r}{r_{CTN}} = V_g - \frac{K_4}{\varepsilon_{ox}} \ln \frac{r_{box}}{r}$$

$$F_{O1}(r) = -\frac{K}{\varepsilon_{ox}} \frac{1}{r}$$
$$F_{n1}(r) = -\frac{K_1}{\varepsilon_n} \frac{1}{r} - \frac{q n_{t\_tn1}}{2\varepsilon_n} r$$
$$F_{O2}(r) = -\frac{K_2}{\varepsilon_{ox}} \frac{1}{r}$$
$$F_{CTN}(r) = -\frac{K_3}{\varepsilon_n} \frac{1}{r} - \frac{q n_{t\_CTN}}{2\varepsilon_n} r$$
$$F_{box}(r) = -\frac{K_4}{\varepsilon_{ox}} \frac{1}{r}$$

$$K = \frac{V_g}{\alpha} - \frac{q n_{t\_tn1}(r_{n1}^2 - r_{O1}^2)}{2\alpha} \left\{ -\frac{r_{O1}^2}{r_{n1}^2 - r_{O1}^2} \frac{1}{\varepsilon_n} \ln \frac{r_{n1}}{r_{O1}} + \frac{1}{2\varepsilon_n} + \frac{1}{\varepsilon_{ox}} \ln \frac{r_{tox}}{r_{n1}} + \frac{1}{\varepsilon_n} \ln \frac{r_{CTN}}{r_{tox}} + \frac{1}{\varepsilon_{ox}} \ln \frac{r_{box}}{r_{CTN}} \right\} - \frac{q n_{CTN}(r_{CTN}^2 - r_{tox}^2)}{2\alpha} \left\{ -\frac{r_{tox}^2}{r_{CTN}^2 - r_{tox}^2} \frac{1}{\varepsilon_n} \ln \frac{r_{CTN}}{r_{tox}} + \frac{1}{2\varepsilon_n} + \frac{1}{\varepsilon_{ox}} \ln \frac{r_{box}}{r_{CTN}} \right\}$$
$$K_1 = K - \frac{1}{2} q_{t\_tn1} r_{O1}^2, \quad K_2 = K + \frac{1}{2} q_{t\_tn1} (r_{tn1}^2 - r_{O1}^2)$$
$$K_3 = K + \frac{1}{2} q_{t\_tn1} (r_{tn1}^2 - r_{O1}^2) - \frac{1}{2} q_{t\_CTN} r_{tox}^2, \quad K_4 = K + \frac{1}{2} q_{t\_tn1} (r_{tn1}^2 - r_{O1}^2) + \frac{1}{2} q_{t\_CTN} (r_{CTN}^2 - r_{tox}^2)$$
$$\alpha = \frac{1}{\varepsilon_{ox}} \ln \frac{r_{o1}}{r_o} + \frac{1}{\varepsilon_n} \ln \frac{r_{n1}}{r_{o1}} + \frac{1}{\varepsilon_{ox}} \ln \frac{r_{tox}}{r_{n1}} + \frac{1}{\varepsilon_n} \ln \frac{r_{ctn}}{r_{tox}} + \frac{1}{\varepsilon_{ox}} \ln \frac{r_{box}}{r_{ctn}}$$

Emission rate 모델링 결과식 (Jin, TED 2025)

$$e_{\mathrm{TBT}} = \nu_{\mathrm{TBT}} \times |\mathrm{TC}|$$
$$|\mathrm{TC}| = \prod_{o1,n1} \exp\left(-\frac{4\sqrt{2m_{\mathrm{ox},n}^*}}{3\hbar q} \cdot \frac{\Phi_H^{1.5} - \Phi_L^{1.5}}{F_{\mathrm{eq}}} \right)$$
$$\nu_{\mathrm{TBT}} = \int W(E) \cdot D(E) \cdot (1 - f_r(E)) dE = \frac{2\pi}{\hbar} |\Delta E_T|^2 V_T \times \frac{\sqrt{2m_n^* m_0}}{\pi^2 \hbar^3} \sqrt{E_{\mathrm{trap}} - E_{C,r}}$$

Jin, TED 2025

CTN Trapped Electron Detrapping

BETOX detrapping 연구를 통해 local trap-to-band emission rate 모델링 구축

실제 retention loss 예측을 위해서는 CTN trapped electrons의 trap-to-band detrapping emission 모델링 구축이 필요함

기존모델 (Amoroso, TED 2011)은 CTN charge를 uniform distribution으로 가정하는 한계가 있음

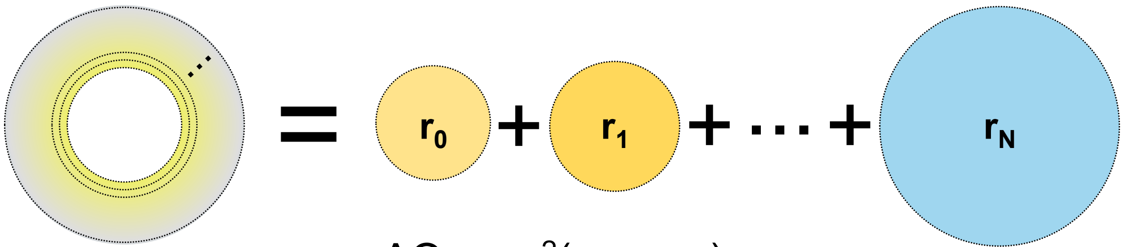
본 연구에서는 arbitrary radial charge distribution에 대한 모델링으로 확장

Jin, EDL 2026

Visualization of Charge Node Analysis

CTN 영역을 N개의 radial charge cell으로 분할

$$Q = \Delta Q_0 + \Delta Q_1 + \cdots + \Delta Q_N$$

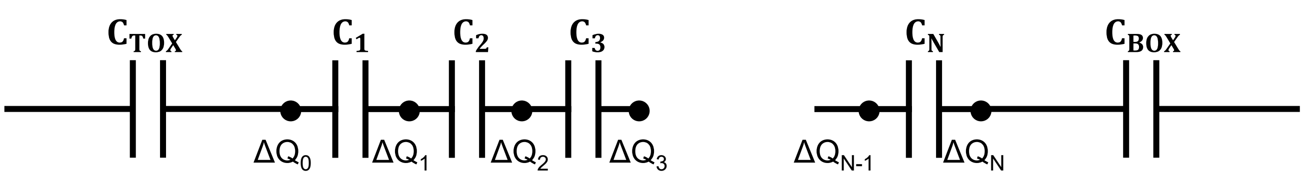


CTN charge

$$\Delta Q_i = \pi r_i^2 (n_{t,i+1} - n_{t,i})$$

$$\text{where, } i=1\text{-}N \begin{cases} n_{t,0} = 0 \\ n_{t,N+1} = 0 \end{cases}$$

각각의 equivalent charge node ΔQ_i으로 표현할 수 있음



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Detailed Poisson Solution

N+2 layer 의 전위 분포

$$\begin{cases} V_{\mathrm{Si}}(r_{\mathrm{Si}}) = V_{\mathrm{ch}} \\ V_{\mathrm{TOX}}(r) = V_{\mathrm{ch}} + \frac{K_{\mathrm{TOX}}}{\epsilon_{\mathrm{TOX}}} \ln \frac{r}{r_{\mathrm{Si}}} \\ V_{\mathrm{CTN},i}(r) = V_{\mathrm{CTN},i-1}(r_{i-1}) + \frac{K_i}{\epsilon_{\mathrm{CTN}}} \ln \frac{r}{r_{i-1}} \\ V_{\mathrm{BOX}}(r) = V_{\mathrm{CTN},N}(r_N) + \frac{K_{\mathrm{BOX}}}{\epsilon_{\mathrm{BOX}}} \ln \frac{r}{r_N} \\ V_{\mathrm{BOX}}(r_{\mathrm{BOX}}) = V_{\mathrm{G}} \end{cases}$$

선형방정식

$$A\mathbf{x} = \mathbf{b}$$

$$\mathbf{x} = \begin{bmatrix} K_{TOX} & K_1 & \cdots & K_N & K_{BOX} \end{bmatrix}^\top$$

경계조건 행렬식 정리

$$A = \begin{bmatrix} 1 & -1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & -1 & \cdots & 0 & 0 \\ \vdots & & & \ddots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 & -1 \\ \frac{2\pi}{C_{\mathrm{TOX}}} & \frac{2\pi}{C_1} & \frac{2\pi}{C_2} & \cdots & \frac{2\pi}{C_N} & \frac{2\pi}{C_{\mathrm{BOX}}} \end{bmatrix}$$

$$\mathbf{b} = \begin{bmatrix} \frac{\Delta Q_0}{2\pi} & \frac{\Delta Q_1}{2\pi} & \cdots & \frac{\Delta Q_N}{2\pi} & \Delta V \end{bmatrix}^\top$$

$$\Delta Q_i = \pi r_i^2 \left(n_{t,i} - n_{t,i-1}\right)$$

$$\Delta V = (V_{\mathrm{G}} - V_{\mathrm{ch}}) - \sum_{i=1}^{N} \frac{q n_{t,i}}{4\epsilon_{\mathrm{CTN}}} \left(r_i^2 - r_{i-1}^2\right)$$

역행렬 계산…?

$\mathbf{x} = A^{-1}\mathbf{b}$ ..? $\longrightarrow$ ΔV_{th}를 구하는데 있어 모든 coefficient 계산은 불필요!

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Q-ΔV_{th} Optimized Calculation

ΔV_{th} : channel에서의 전기장의 세기가 중요

$$F_{\mathrm{TOX}}(r_{\mathrm{Si}}) = \frac{K_{\mathrm{TOX}}}{\epsilon_{\mathrm{TOX}}} \frac{1}{r_{\mathrm{Si}}}$$

K_{TOX} 값을 일치시키는게 ΔV_{th} 값 구하는 핵심 과정!

Cramer's rule에 따르면,

$$K_{\mathrm{TOX}} = \frac{\det\left[A_{\mathbf{b}(\Delta Q_i, \Delta V_{\mathrm{th}})}\right]}{\det A}$$

$$\det\left[A_{\mathbf{b}(0, \Delta V_{\mathrm{th}})}\right] = \det\left[A_{\mathbf{b}(\Delta Q_i, 0)}\right]$$

$$\det \begin{bmatrix} 0 & -1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & -1 & \cdots & 0 & 0 \\ \vdots & & & \ddots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 & -1 \\ \Delta V_{\mathrm{th}} & \frac{2\pi}{C_1} & \frac{2\pi}{C_2} & \cdots & \frac{2\pi}{C_N} & \frac{2\pi}{C_{\mathrm{BOX}}} \end{bmatrix} = -\Delta V_{\mathrm{th}}$$

$$\therefore \Delta V_{\mathrm{th}} = -\det\left[A_{\mathbf{b}(\Delta Q_i, 0)}\right]$$

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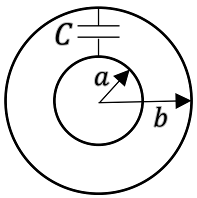
Q-ΔV_{th} Optimized Calculation

결론: 임의로 주어진 CTN내의 discretized 분포에 대한 ΔV_{th} 계산식

초기 입력 조건: $\Delta Q_i \ (i = 1 - N),\ C = \dfrac{2\pi\varepsilon}{\ln(b/a)}$



$$\therefore\ \Delta V_{th} = -\det\left[A_{b(\Delta Q_i,0)}\right] = -\det\begin{bmatrix} \frac{\Delta Q_0}{2\pi} & -1 & 0 & \cdots & 0 & 0 \\ \frac{\Delta Q_1}{2\pi} & 1 & -1 & & 0 & 0 \\ & \vdots & & \ddots & & \vdots \\ \frac{\Delta Q_N}{2\pi} & 0 & 0 & \cdots & 1 & -1 \\ 0 & \frac{2\pi}{C_1} & \frac{2\pi}{C_2} & & \frac{2\pi}{C_N} & \frac{2\pi}{C_{BOX}} \end{bmatrix}$$

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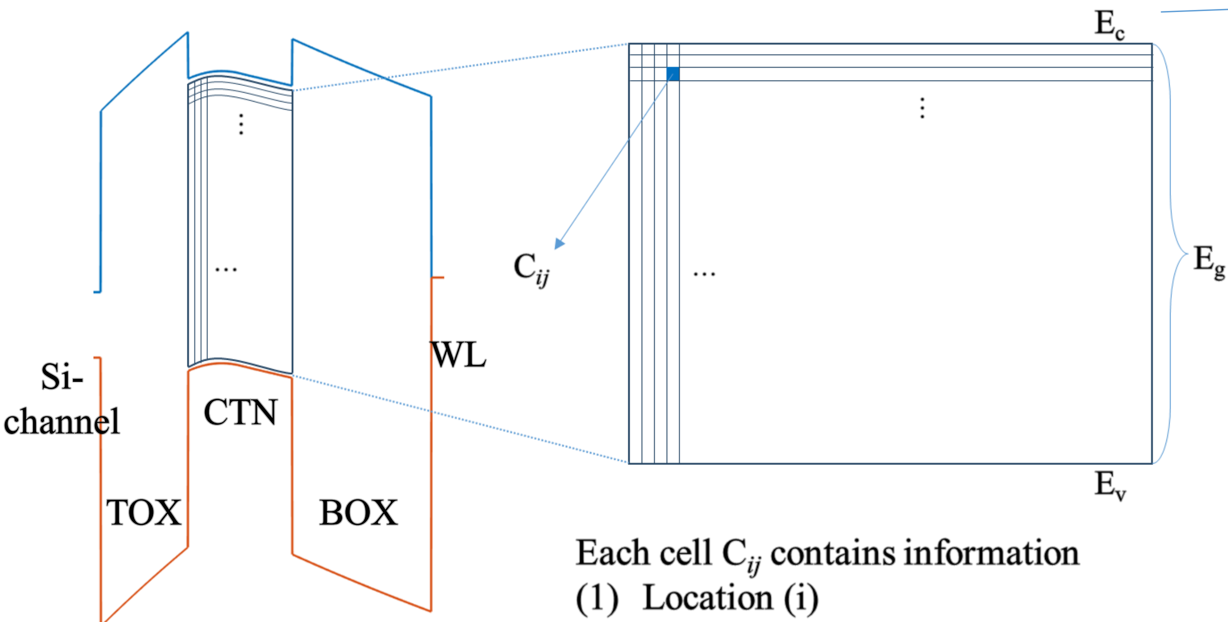
Time-step-free Physics-based Modeling

핵심 원리: initial trapped charge distribution으로 부터 target retention time의 ΔV_{th}를 직접 계산



Each cell C_{ij} contains information
(1) Location (i)
(2) Trap depth (j)
(3) Trap concentration
(4) Emission rate to channel direction
(5) Emission rate to wordline direction

1. Initial state
$n_{t0}(r, E_t)$

2. Emission rates
e_{ch}, e_{WL}

3. Direct evaluation
$n_t(r, E_t, t)$

계산 비용은 final retention time 자체가 아니라 trap bin 수와 evaluated time point 수에 의해 결정

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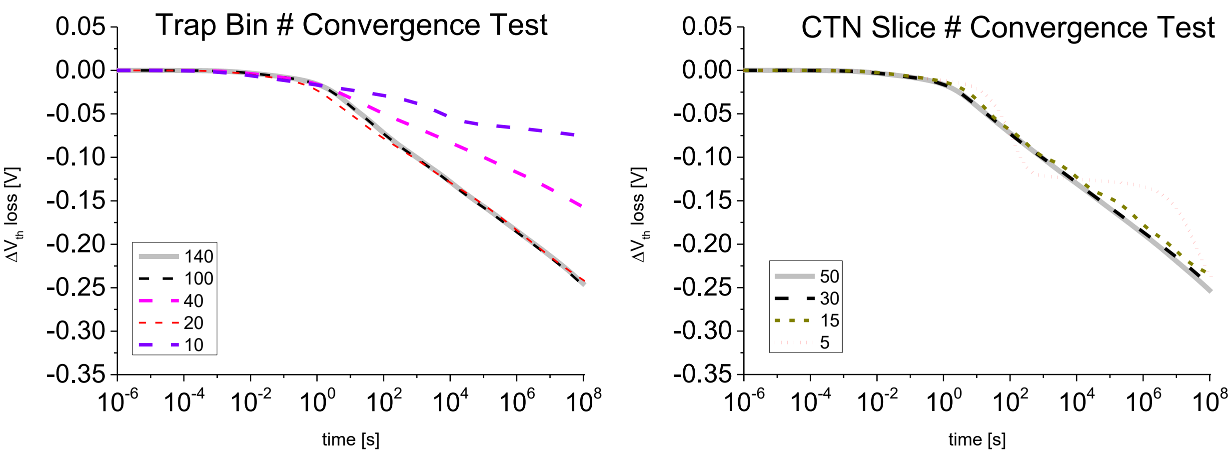
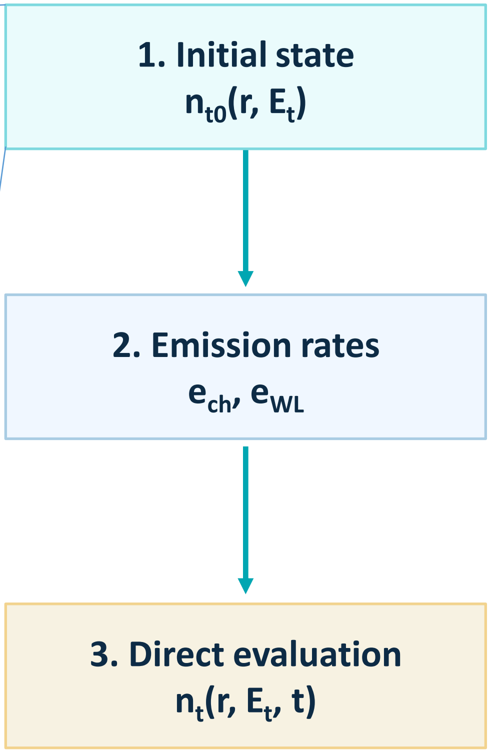
Time-step-free Physics-based Modeling

핵심 원리: initial trapped charge distribution으로 부터 target retention time의 ΔV_{th}를 직접 계산

- Numerical 상관관계 실험





- trap bin을 10-140 등분하여 비교 테스트
- CTN slice를 5-50 등분하여 비교 테스트
- 100등분 이상부터 수렴
- 30등분 이상부터 수렴

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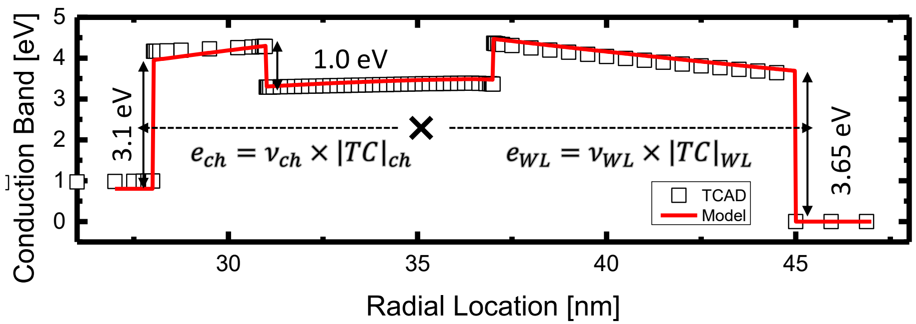
Time-step-free Physics-based Modeling

핵심 원리: initial trapped charge distribution으로 부터 target retention time의 ΔV_{th}를 직접 계산



주어진 initial band diagram을 기준으로 channel 방향과 word line (WL) 방향으로의 emission rate을 계산

$$e = \nu \times |TC|$$

$$\nu = \frac{2\pi}{\hbar} |E_t|^2 V_{\text{trap}} \cdot \frac{m_0\sqrt{2m^*}}{\pi^2\hbar^3} \sqrt{E - E_{\text{trap}}}$$

$$|TC| = \prod \exp\left(-\frac{4\sqrt{2m^*}}{3\hbar q} \frac{\Phi_H^{1.5} - \Phi_L^{1.5}}{F}\right)$$

1. Initial state
$n_{t0}(r, E_t)$

↓

2. Emission rates
e_{ch}, e_{WL}

↓

3. Direct evaluation
$n_t(r, E_t, t)$

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Time-step-free Physics-based Modeling

핵심 원리: initial trapped charge distribution으로 부터 target retention time의 ΔV_{th}를 직접 계산

$$\frac{d}{dt}x(t) = f(x(t)) \quad \text{: Autonomous ODE}$$

Detrapping Autonomous ODE

$$\frac{d}{dt}n_t(r, E_t, t) = -(e_{\text{ch}} + e_{\text{WL}})n_t(r, E_t, t) \qquad (12)$$

위 식을 풀면, 시간에 따른 일반해를 구할 수 있음

$$\boxed{n_t(r, E_t, t) = \sum_r \sum_{E_t} n_{t0}(r, E_t) \exp(-(e_{\text{ch}} + e_{\text{WL}})t) \qquad (13)}$$

초기 전하 분포

closed-form evaluation

1. Initial state
$n_{t0}(r, E_t)$

2. Emission rates
e_{ch}, e_{WL}

3. Direct evaluation
$n_t(r, E_t, t)$

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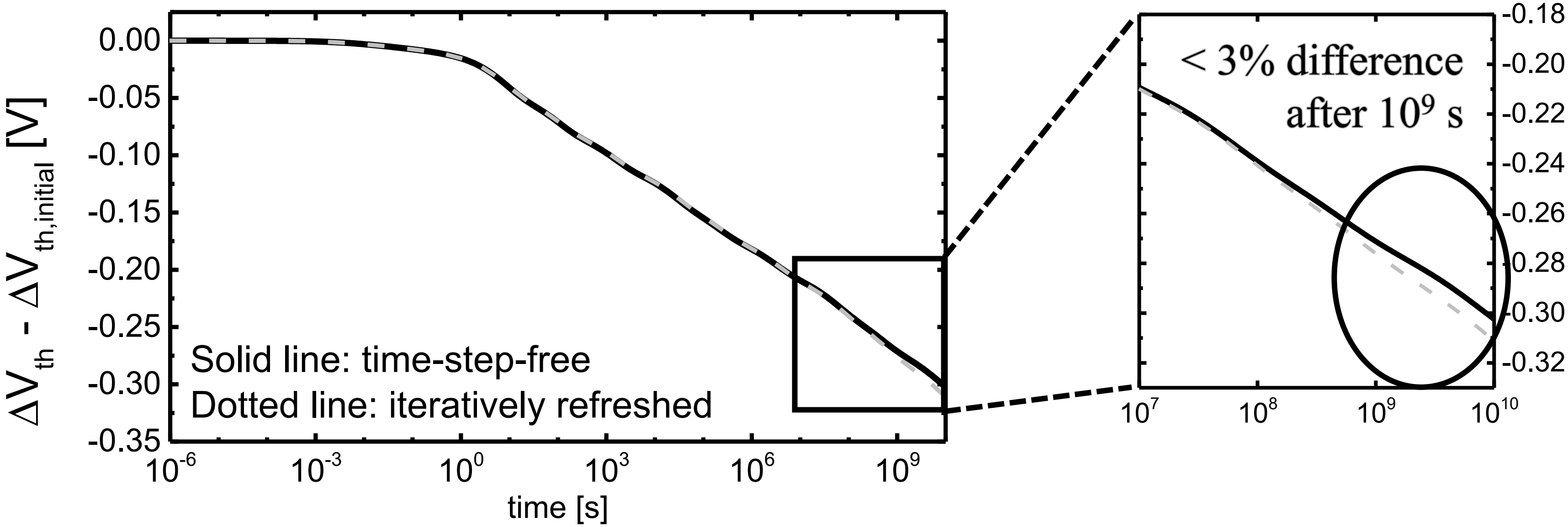
Validation of Time-Step-Free Evaluation

Time-step-free evaluation의 validity를 확인하기 위해 iteration test를 수행함



Iterative test를 진행한 결과와 iteration이 없는 단일계산으로 진행한 time-step-free evaluation의 차이가 10^9 [s] 이상의 시간이 지나야 3%가량 차이 발생

두 결과의 차이가 작아 time-step-free approximation이 유효함을 확인함

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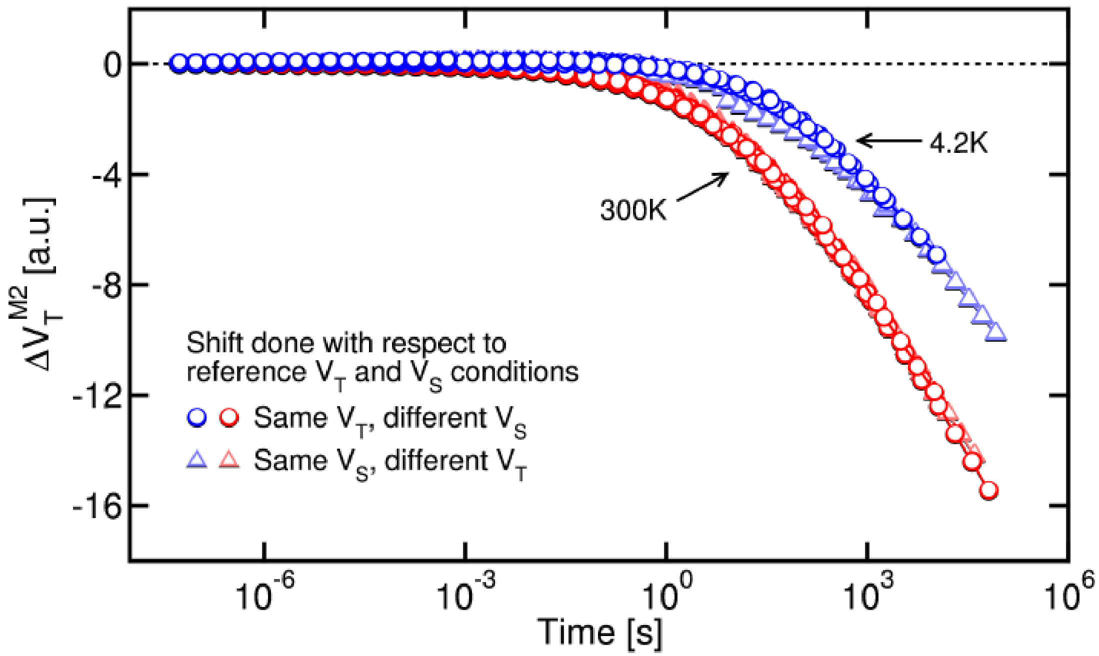
Cryogenic Retention Condition

극저온 조건이 필요한 이유?



실험 측정 결과
(Refaldi, IRPS 2025)

- 실험적으로 Detrapping effect를 분리하는 기법
  - Thermal transition 억제
  - Detrapping effect 분리

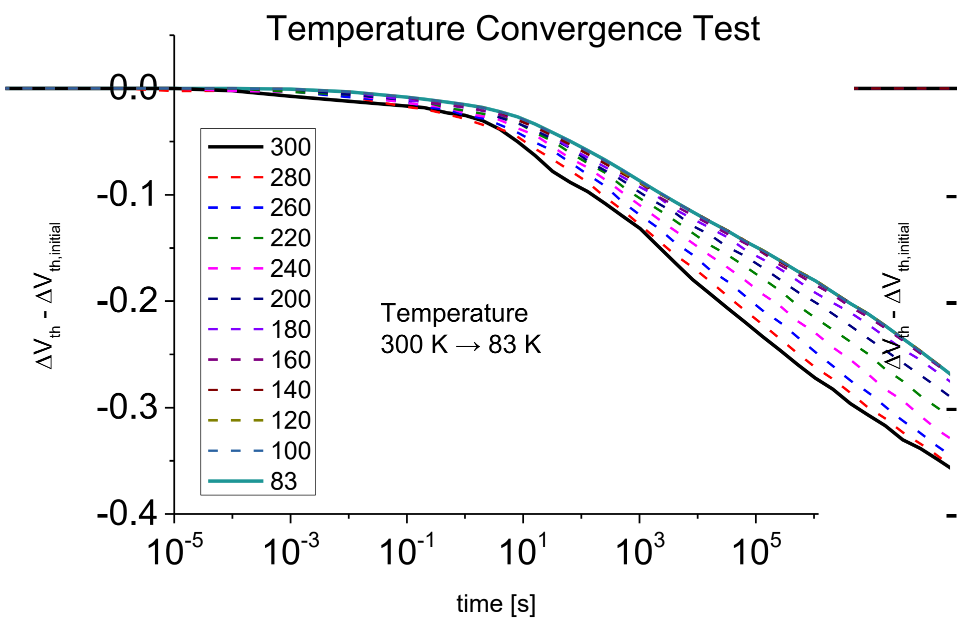
- Modeling validation에 적합한 measurement reference 제공 (단, 단위는 A.U.)

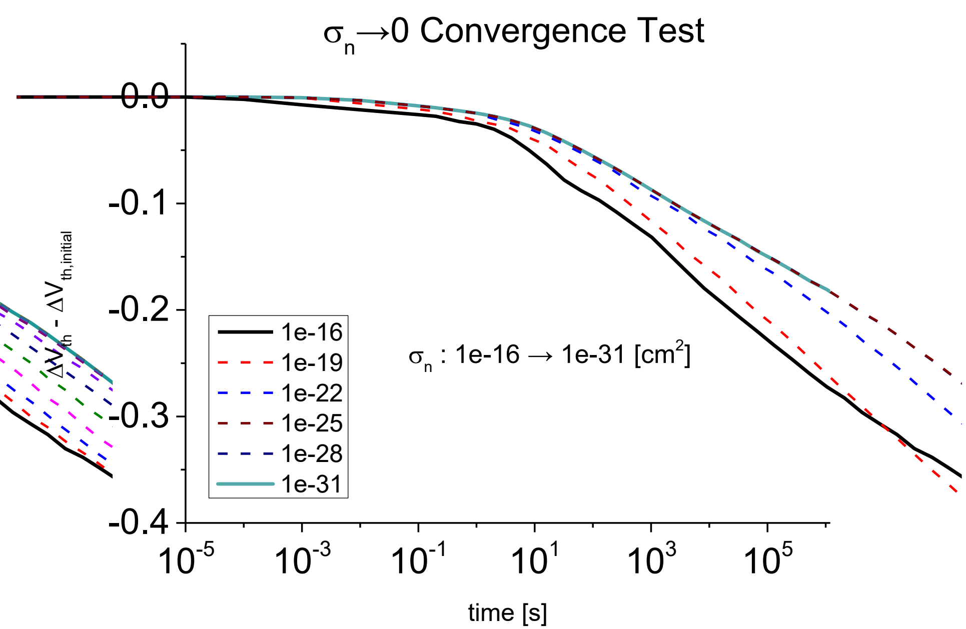
Cryogenic-like TCAD Emulation

온도 – σ_n 수렴성 테스트: 극저온 환경과 $\sigma_n \rightarrow 0$ 사이의 상호

Cyrogenic-like TCAD Emulation Physics deck

- SRH Recombination, Fermi statistics, non-local tunneling, Poole-Frenkel emission, thermionic emission





300 K → 83 K (TCAD 최저온도)

$10^{-16} \rightarrow 10^{-31} \rightarrow \ldots$ [cm^2]

$\sigma_n \rightarrow 0$ 지속적으로 확장 가능

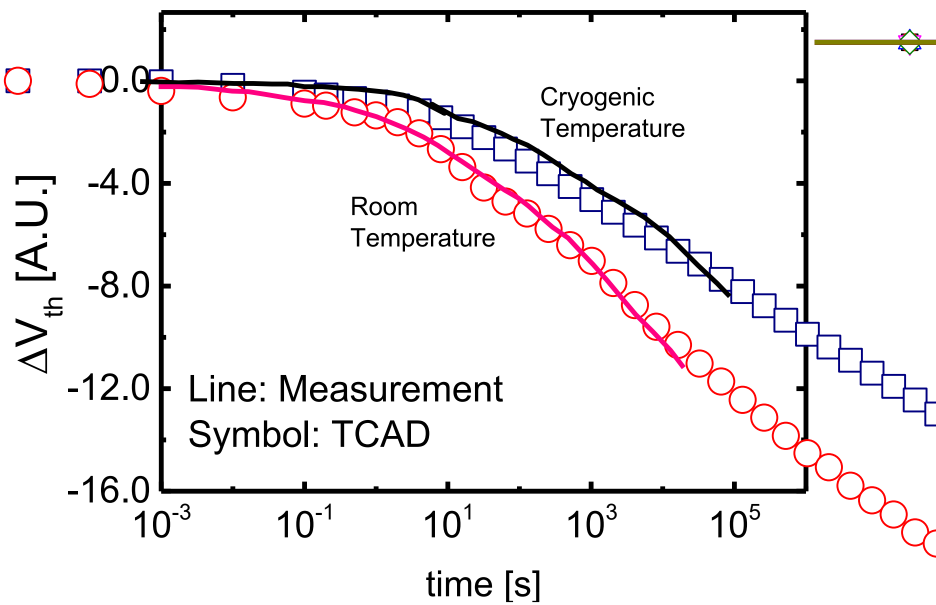
Cryogenic-like Emulation

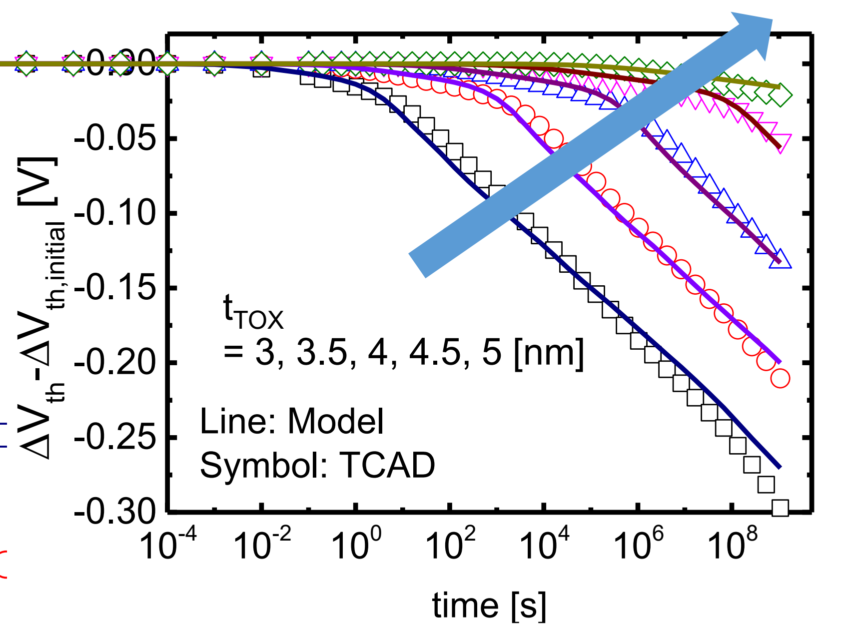
측정치 – TCAD

- 측정치와 TCAD calibration 진행
- ΔV_{th}는 A.U. 값이지만, t_{TOX} 값 조절을 하며 time scale을 기준으로 calibration

모델 – TCAD

- t_{TOX} = 3 ~ 5 [nm] 변화에 따른 변동 검증





- O/N/O = **3**/6/8 [nm]

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Computational Impact

Property	Transient TCAD	Time-step-free model
Temporal treatment	Sequential time stepping	Closed-form evaluation of Eq. (13)
Final-time dependence	Runtime grows as simulated retention time increases	**Independent of final physical time for fixed queries**
Average runtime	**27483 s**	**3.926 s**

≈ 7000×

runtime reduction

- TCAD 는 final physical time까지 time step 별 순차적으로 계산
- Time-step-free model은 closed-form evaluation
- Computational resource는 final retention time 자체가 아니라 trap bin 수와 evaluated time point 수에 의해 결정

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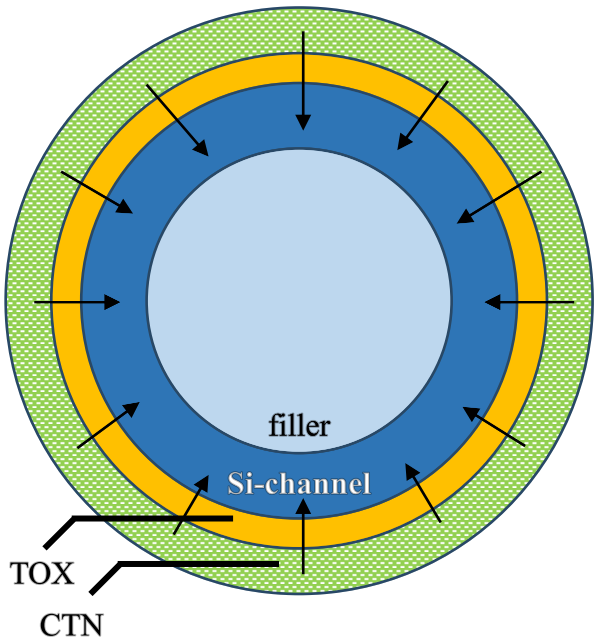
Limited Contribution of TAT

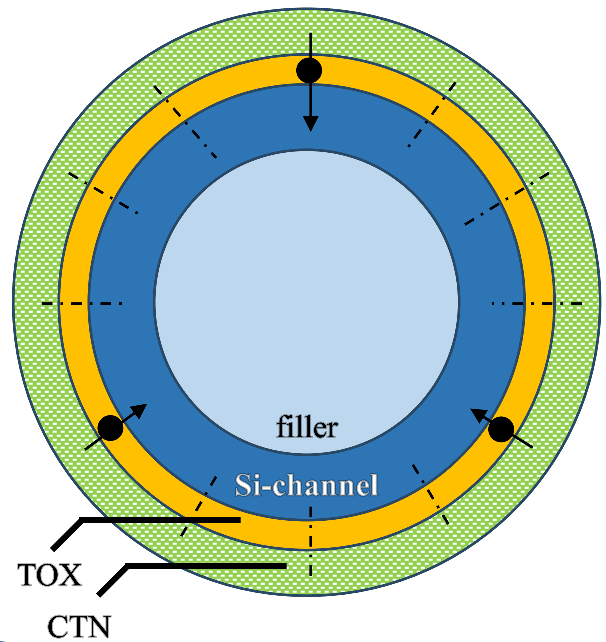
가능 경로 수 (Route Count) 개념 활용하여 TAT 제외 이유

CTN > Si-channel (Detrap)



가능 경로 수 (Route Count) ➡ detrap 통과 넓이 $2\pi r_{TOX,\ centroid} L_{WL}$

CTN > TOX > Si-channel (TAT)

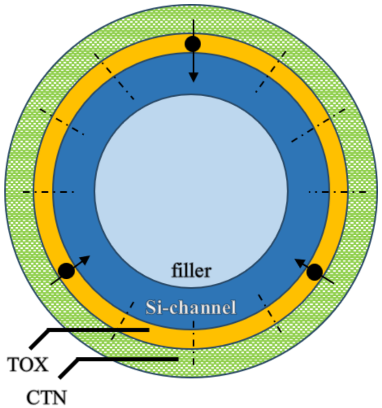


가능 경로 수 (Route Count) ➡ TAT 통과 넓이 $\#_{trap}\ a_{trap}^{2}$

cf) $a_{trap} \sim 1.7$Å

통과 넓이 대폭 감소로 인한 가능 경로 수 감소

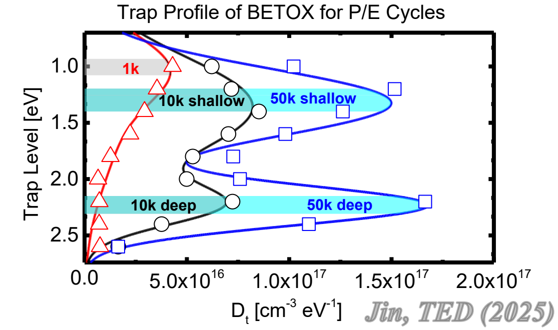
Limited Contribution of TAT



CTN → TOX → Si-channel

$$\nu_{CTN\to TOX} = \frac{2\pi}{\hbar}|E_T|^2 V_{trap} \times D_t$$

Detrapping Model *Jin, TED (2025)* *Jin, EDL (2026)*



Jin, TED (2025)

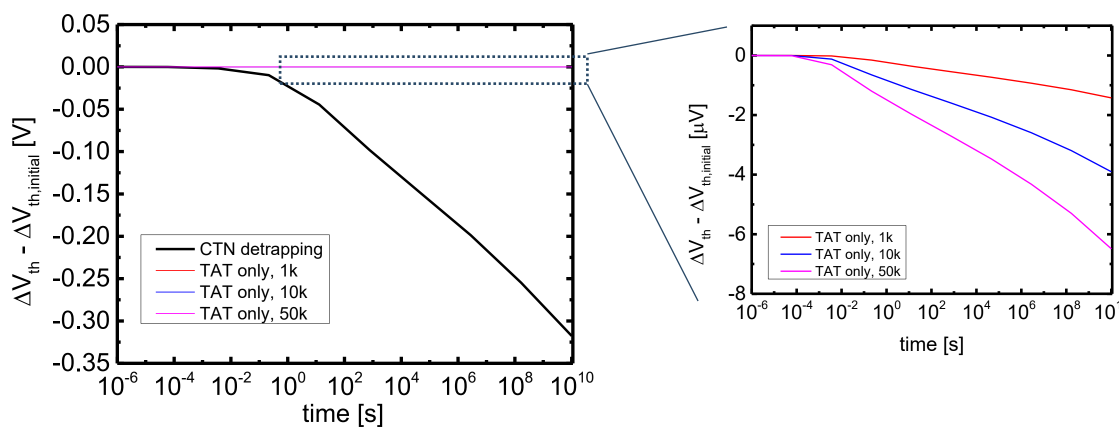
Effective route count modeling

$$e_{CTN\to TOX} = \nu \times |TC| \times \frac{\#_{trap}\, a_{trap}^2}{2\pi r_{TOX,centroid} L_{WL}}$$

BETOX total trap density (measurement)	CTN trap [#]
1k: 4.49×10^{16} cm^{-3}	0.55
10k: 1.16×10^{17} cm^{-3}	1.42
50k: 1.89×10^{17} cm^{-3}	2.03

∴ Simulation Result

endurance 별로 차이는 있으나, CTN detrap의 넓은 emission route를 극복할 순 없다



Summary and Future Works

Summary

- BETOX trap-to-band tunneling emission rate 모델링 (Jin, TED 2025)
- CTN 전자 임의 분포에 대한 ΔV_{th} 계산 framework 구축
- Closed-form autonomous ODE 기반 time-step-free retention 모델 제안
- Cryogenic-like TCAD 검증 및 ~7000x 시뮬레이션 시간 단축 확인
- 현 모델 기반으로 detrap 성분과 TAT 성분 비교

Future Works

- cell-to-cell variation 및 retention 산포 적용 가능한 모델링 셋업
- SPICE 등 cell 단위의 reliability simulation과의 연계